
LatentGoalOps: A Benchmark for Latent Objective Adaptation Under Silent Shifts

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Latent objectives are common in real operational settings: stakeholders reveal con-
2 straints, preferences, and feedback over time, while the true utility function gov-
3 erning good decisions may remain implicit or change without announcement. We
4 introduce **LatentGoalOps**, a benchmark for evaluating latent objective inference
5 and adaptation under silent objective shifts. Agents observe a fixed public oper-
6 ating schema consisting of business state, stakeholder messages, decision ledgers,
7 delayed effects, and governance constraints, but never observe the hidden objec-
8 tive used by the deterministic grader. The benchmark contains three sequential
9 business-operations environments; our main analysis focuses on two week-long
10 anchor tasks, with a shorter planning task used as breadth evidence. Across the
11 anchor tasks, restricted-view leakage audits show no easy shortcut, models incur
12 a persistent penalty on silently shifted episodes, and the best current local model
13 remains roughly 0.20 below a myopic oracle that observes the active hidden objec-
14 tive but does not solve the full future trajectory. We further find that performance is
15 mechanism-sensitive: several sharp heldout drops are caused by structured-action
16 failures rather than latent-policy quality alone. For this reason, LatentGoalOps re-
17 ports strict score, native-only score, and interface reliability jointly. These results
18 position LatentGoalOps as a benchmark for studying hidden-objective adaptation
19 under a fixed public schema and deterministic grader, rather than as a measure of
20 general business-agent readiness. Small hosted-model probes show heterogeneous
21 interface reliability, but none closes the oracle gap.

22 1 Introduction

23 Many agent benchmarks make the objective explicit: the instruction is written down, the tool con-
24 tract is visible, and success is defined against an external goal. Recent suites such as τ -bench and
25 BrowserGym have pushed that setting toward longer-horizon, tool-using evaluation in realistic inter-
26 faces [Yao et al., 2024, Le Sellier De Chezelles et al., 2024]. Real operating environments, however,
27 often require a different capability. Leadership priorities are not always fully stated, stakeholder
28 messages are noisy, visible KPI movement is ambiguous, and the underlying objective can change
29 without announcement.

30 This paper targets that gap. We study *latent objective inference and adaptation*: inferring an unstated
31 reward structure from public evidence, acting coherently through a strict interface, and adapting
32 when the hidden objective changes mid-episode. We instantiate this construct in **LatentGoalOps**,
33 a synthetic but structured operating-world benchmark with persistent entities, delayed effects, de-
34 terministic grading, and silent objective shifts. The synthetic construction is intentional: it gives
35 the benchmark hidden-objective ground truth while preserving a realistic operating surface for the
36 agent.

37 The release contains three paper-facing sequential environments: *P1 Startup Week*, *P2 Incident*
38 *Response Week*, and *P3 Quarterly Headcount Plan*. The paper centers P1 and P2 as anchor tasks; P3
39 serves only as shorter breadth evidence. Within that scope, the frozen evidence supports three claims:
40 current local open-weight configurations retain a clear myopic-oracle gap, within-family behavior is
41 heterogeneous rather than uniformly monotone, and some large heldout drops are interface-mediated
42 rather than pure policy regressions.

43 **Related work and positioning.** Existing interactive benchmarks such as τ -bench and Browser-
44 Gym primarily evaluate explicit user goals in realistic interfaces, while structured-output and
45 function-calling evaluations such as JSONSchemaBench and BFCL isolate interface control itself
46 [Yao et al., 2024, Le Sellier De Chezelles et al., 2024, Geng et al., 2025, Patil et al., 2025]. La-
47 tentGoalOps instead hides the objective, introduces delayed effects and silent shifts, and requires
48 strict structured actions under non-stationarity. Following recent benchmark-methodology work, we
49 therefore report strict score, native-only score, and interface reliability together rather than a single
50 leaderboard scalar [Authors, 2025, Reuel et al., 2024, Mazumder et al., 2023, Longjohn et al., 2024].

51 Contributions.

- 52 • We introduce LatentGoalOps, a benchmark suite for latent objective inference and adaptation un-
53 der silent non-stationarity with delayed effects, hidden strategic objectives, deterministic grading,
54 two week-long anchor environments, and one shorter breadth task.
- 55 • We release a runnable benchmark artifact with a frozen protocol, core and heldout splits, deter-
56 ministic baselines, restricted-view leakage probes, generated paper assets, and reviewer-facing
57 reproducibility documentation.
- 58 • We report a strengthened local benchmark matrix with 630 primary model episodes and 360 base-
59 line episodes, plus a P1/P2 anchor-task robustness slice that isolates the benchmark’s strongest
60 latent-adaptation settings.
- 61 • We add exploratory hosted-model probes for Gemini 3 Flash preview, GPT-5.4 mini, and Claude
62 Sonnet 4.6 to test whether accessible API models immediately close the myopic-oracle gap under
63 the same action shell.
- 64 • We show that the benchmark’s empirical story is mechanism-sensitive: some larger-model
65 changes are native-policy movements, while some of the sharpest heldout regressions are
66 interface-mediated drops.

Benchmark	Business ops	Hidden objective	Silent shift	Delayed effects	Strict schema
WebArena	No	No	No	No	No
WorkArena	Yes	No	No	No	No
BrowserGym	No	No	No	No	No
OSWorld	No	No	No	No	No
τ -bench	Yes	No	No	Limited	No
LatentGoalOps	Yes	Yes	Yes	Yes	Yes

Table 1: Qualitative construct comparison against nearby interactive-agent benchmarks [Zhou et al., 2023, Drouin et al., 2024, Le Sellier De Chezelles et al., 2024, Xie et al., 2024, Yao et al., 2024]. LatentGoalOps differs by combining hidden objective, silent shift, delayed effects, and a strict executable action shell.

2 Benchmark

2.1 Environment Loop

Each episode is a finite partially observed decision process. At step t , the agent receives a public observation o_t , emits a structured action a_t , and the environment advances to the next state. Public observations include inbox messages, KPI dashboards, dated decision ledgers, pending effects, realized effects, customer accounts, stakeholder personas, team state, market context, and visible governance constraints. The hidden objective g_t is never shown directly.

Formally, an episode is a trajectory

$$\tau = (o_0, a_0, o_1, a_1, \dots, o_T),$$

with hidden state x_t , public observation $o_t = \psi(x_t)$, and active hidden objective state g_t . The environment maintains a normalized four-channel metric vector $m_t \in [0, 1]^4$ over growth, retention, revenue, and efficiency, and an active latent weight vector $w_t \in \Delta^4$. The latent utility used by the grader is

$$U_t = \text{clip}_{[0,1]} \left(\sum_{c \in \mathcal{C}} m_{t,c} w_{t,c} \right), \quad \mathcal{C} = \{\text{growth, retention, revenue, efficiency}\}.$$

The action interface is intentionally strict. On the sequential tasks, a valid action is a JSON-serializable payload over task-specific fields such as `chosen_initiatives`, `messaging_action`, `pricing_change_pct`, `support_policy`, or `budget_allocations`. If the model fails to emit a valid structured action, the benchmark executes a fallback action and the trajectory still advances. In other words, interface failures are part of the task, not merely parser trivia. This follows the broader lesson from structured-generation and tool-use evaluation that output control is part of agent capability rather than an implementation footnote [Geng et al., 2025, Patil et al., 2025].

2.2 Three Sequential Environments and Silent Shifts

LatentGoalOps evaluates three sequential operating environments. The implementation preserves internal task ids for reproducibility, but the main paper uses the cleaner P1–P3 names throughout. Table 2 summarizes the benchmark environments and their roles in the manuscript.

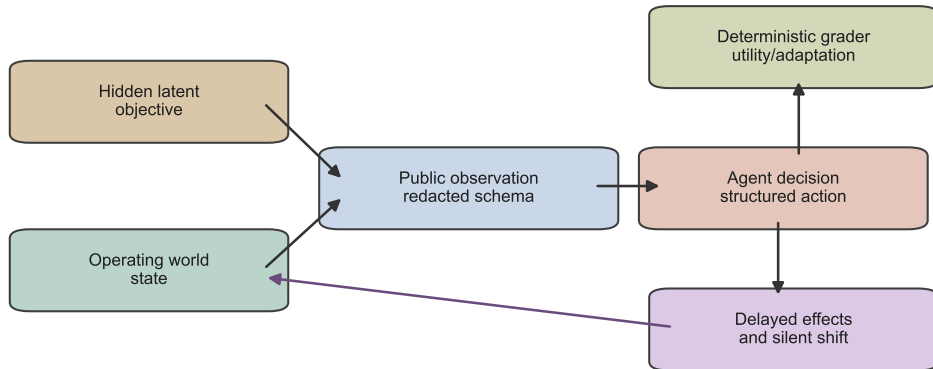
P1 Startup Week is a ten-step operating week with initiatives, renewals, and delayed business consequences. P2 Incident Response Week is a seven-step incident setting where trust repair, governance, recovery speed, and cost can conflict. P3 Quarterly Headcount Plan is a shorter staffing task used only for breadth. Heldout evaluation changes scenario families and delayed-effect realizations while preserving the same public schema and grading semantics. Silent shifts occur with probability 0.6, usually at steps $\{3, 4, 5\}$, and can alter latent weights as well as risk posture, planning horizon, segment focus, and governance strictness. The model is never told that the shift occurred.

2.3 Scoring and Reporting

For the primary sequential tasks, the episode score is an explicit weighted combination of final latent utility U , adaptation A , coherence C , and constraint satisfaction K :

$$S_{\text{task}} = \alpha_U U + \alpha_A A + \alpha_C C + \alpha_K K.$$

LatentGoalOps evaluates whether agents infer and adapt to hidden business objectives



The agent only sees public observations. The latent objective and delayed consequences shape future rewards.

Figure 1: LatentGoalOps exposes public operating state and delayed consequences while hiding the objective that determines utility. The agent must act through a strict structured-action interface; failures at that interface alter the trajectory itself.

Paper ID	Environment	Horizon	Paper role	Hidden shift
P1	Startup Week	10	anchor	Yes
P2	Incident Response Week	7	anchor	Yes
P3	Quarterly Headcount Plan	4	breadth	Yes

Table 2: LatentGoalOps benchmark environments. The main paper’s result-bearing analyses center the two week-long anchor tasks P1 and P2.

100 Task-specific coefficients and no-shift renormalization rules are fixed in the release protocol. This
 101 keeps the main paper focused on the measurement construct rather than coefficient bookkeeping.

102 For *Startup Week* and *Incident Response Week*, adaptation measures how quickly post-shift decisions
 103 improve under the active hidden objective and whether that recovery is sustained; P3 uses the same
 104 principle with a shorter horizon. We report overall strict score, native-only score, shifted versus
 105 unshifted score, and interface reliability. Release aggregates pool 10 seeds per task across P1–P3,
 106 while P1/P2 anchor slices isolate the strongest construct signal; aggregate views use 95% bootstrap
 107 confidence intervals.

108 3 Evaluation and Benchmark Validity

109 3.1 Baseline Separation

110 LatentGoalOps has meaningful score dynamic range on the frozen release aggregate over P1–P3.
 111 Random, visible-heuristic, and myopic-oracle baselines preserve the expected ordering on both
 112 splits, and the myopic-oracle gap remains substantial on both splits. The myopic oracle observes
 113 the active hidden objective and chooses the best immediate action, but does not solve the full future
 114 trajectory; we use it as a construct headroom reference rather than as a fair agent baseline or human
 115 proxy.

116 3.2 Leakage Audit

117 Restricted-view centroid probes test for shortcut access to the hidden objective. Table 4 shows
 118 that P1 and P2 stay near the four-way chance regime under both full and restricted views, with
 119 `explicit_goal_token_rate` equal to 0.0 throughout. P3 is less clean: its core restricted-view
 120 maximum reaches 0.35, which is why we treat it as breadth evidence rather than as a validity anchor.
 121 Each probe slice uses 20 train and 20 test examples, so this audit is a shortcut screen rather than a
 122 proof of impossibility.

Baseline	Split	Episodes	Mean score	95% CI
random	core	60	0.303	[0.286, 0.320]
visible heuristic	core	60	0.346	[0.329, 0.362]
myopic oracle	core	60	0.599	[0.575, 0.623]
random	heldout	60	0.310	[0.290, 0.330]
visible heuristic	heldout	60	0.371	[0.356, 0.388]
myopic oracle	heldout	60	0.598	[0.575, 0.623]

Table 3: Baseline separation on the frozen primary-task aggregate.

Task	Split	Full	Numeric	Sender	Order	Max restricted	Goal tokens
Startup Week	core	0.20	0.05	0.25	0.25	0.30	0.00
Startup Week	heldout	0.20	0.15	0.20	0.25	0.25	0.00
Incident Response Week	core	0.20	0.10	0.30	0.25	0.30	0.00
Incident Response Week	heldout	0.15	0.20	0.30	0.25	0.30	0.00
Quarterly Headcount Plan	core	0.25	0.25	0.25	0.25	0.35	0.00
Quarterly Headcount Plan	heldout	0.20	0.15	0.25	0.25	0.30	0.00

Table 4: Compact restricted-view leakage summary for paper-facing tasks. Chance is 0.25 for the four-way archetype probe. “Max restricted” is the maximum over non-full views in the frozen audit and is interpreted as a screening diagnostic rather than a proof of no shortcut.

Utility-only rescoring sharply compresses the myopic-oracle gap, which is the expected pattern if the benchmark is measuring latent-objective adaptation rather than only final-state utility.

4 Main Results

4.1 The Two Week-Long Anchor Tasks Carry the Main Result

Because P3 is shorter-horizon and less discriminative than the two week-long environments, Table 5 reports a P1/P2-only robustness slice computed directly from the frozen release as the paper’s main result view. On the anchor slice, the strongest local core mean is Qwen 27B at 0.425 and the strongest heldout mean is DeepSeek 8B at 0.431, while the myopic-oracle anchor means are 0.626 on core and 0.630 on heldout. The best local anchor results therefore still trail the oracle by about 0.20.

Taken together with the leakage audit and the persistent shifted-versus- unshifted gap, the P1/P2 anchor tasks make the intended construct visible: public trajectory evidence is informative but incomplete, so successful behavior requires latent-goal tracking rather than shortcut extraction.

4.2 The Full Release Aggregate Preserves the Same Story

The full P1–P3 aggregate preserves the same story: the strongest frozen local aggregates are 0.443 on core and 0.446 on heldout, still about 0.15 below the myopic-oracle baselines of 0.599 and 0.598.

4.3 Hosted API Probes Improve Reliability but Do Not Close the Gap

To test whether readily accessible hosted models close the benchmark’s oracle gap, we ran small OpenRouter probes on the P1–P3 aggregate after freezing the primary local-model matrix. These runs are exploratory and not used as load-bearing within-family scaling evidence, but they answer a practical reviewer question: whether modern API models immediately erase the oracle gap. Because P1 and P2 are the load-bearing anchor tasks, Table 6 reports both P1/P2 means and the broader P1–P3 aggregate.

The probes separate interface reliability from latent-objective quality. Gemini 3 Flash preview often produced plausible intent with non-native field names, yielding low native rates and many empty fallbacks. GPT-5.4 mini was mostly native and reached 0.388/0.352 on the P1/P2 core/heldout anchor slice (0.418/0.381 on P1–P3). Claude Sonnet 4.6 was the cleanest probe, with native rate 1.00 on both splits and core/heldout P1/P2 scores of 0.423/0.385. None of these descriptive point

Model	Core P1/P2	Heldout P1/P2	Core native rate	Heldout native rate
Myopic oracle	0.626	0.630	1.00	1.00
DeepSeek 1.5B	0.407	0.407	1.00	1.00
DeepSeek 8B	0.408	0.431	1.00	1.00
DeepSeek 14B	0.420	0.405	1.00	1.00
Gemma 2B	0.401	0.412	1.00	1.00
Gemma 4B	0.408	0.416	1.00	1.00
Gemma 26B	0.392	0.341	0.65	0.70
GPT-OSS 20B	0.407	0.358	1.00	1.00
Qwen 2B	0.406	0.412	1.00	1.00
Qwen 9B	0.394	0.393	1.00	1.00
Qwen 27B	0.425	0.340	0.95	0.25

Table 5: P1/P2 robustness slice from the frozen release. The myopic-oracle row makes the remaining adaptation headroom explicit, while the weakest heldout cells coincide with shell-fragile models.

Model	Split	Episodes	P1/P2	P1-P3	Native-only	Native n	Native rate	Empty fallback
Gemini 3 Flash preview	core	30	0.346	0.328	0.385	8	0.27	0.73
Gemini 3 Flash preview	heldout	30	0.319	0.313	0.338	6	0.20	0.80
GPT-5.4 mini	core	30	0.388	0.418	0.414	27	0.90	0.10
GPT-5.4 mini	heldout	30	0.352	0.381	0.402	21	0.70	0.30
Claude Sonnet 4.6	core	30	0.423	0.429	0.429	30	1.00	0.00
Claude Sonnet 4.6	heldout	30	0.385	0.393	0.393	30	1.00	0.00

Table 6: Exploratory hosted-model probes run through OpenRouter on the P1-P3 aggregate. These API probes are excluded from the primary local-model matrix and are used only to test whether readily accessible hosted models close the gap to the myopic oracle. Entries are descriptive point estimates.

estimates closes the myopic-oracle gap. Under these small probes, better shell reliability does not by itself close the gap to a privileged policy that sees the active hidden objective.

4.4 Some of the Sharpest Heldout Drops Are Interface-Mediated

Across the frozen release, scaling is family-specific rather than monotone, but the most important mechanism is interface confounding rather than raw ranking alone. DeepSeek remains the cleanest native-policy signal: its three reported sizes stay at native rate 1.0 on both splits, so its gains and losses are best read as policy movement. Qwen 2B $\rightarrow$ 9B also changes at native rate 1.0, giving a clean example of non-monotone native-policy behavior. The sharp heldout failures, by contrast, are dominated by shell fragility.

The clearest examples are Gemma 26B and Qwen 27B. On the P1/P2 anchor slice, Gemma 26B heldout falls to 0.341 overall at native rate 0.70, while Qwen 27B heldout falls to 0.340 overall at native rate 0.25. On the full release aggregate, Qwen 27B heldout rises from 0.385 overall to 0.457 on the native subset, but that subset contains only 14 episodes and must be read diagnostically rather than as a leaderboard claim. Table 7 grounds that mechanism in exact stored outputs: DeepSeek stays incident- focused across the shift, whereas Qwen 27B hits an empty fallback and later returns stale growth/revenue moves.

4.5 Silent Shifts Remain the Dominant Stressor

Silent objective shifts remain the primary benchmark stress test. Across the release matrix, shifted episodes are materially harder than unshifted ones on both core and heldout, with a median unshifted-minus-shifted gap of 0.172 in the release-wide diagnostics. This gap is not explained away by shell failures: models with perfect native-action reliability, including all DeepSeek sizes, still pay a substantial shift penalty. That is the benchmark construct we most want to isolate. Figure 3 makes this visible at the aggregate level.

Observed score vs. native-only score (native-only estimates marked when native n is small)

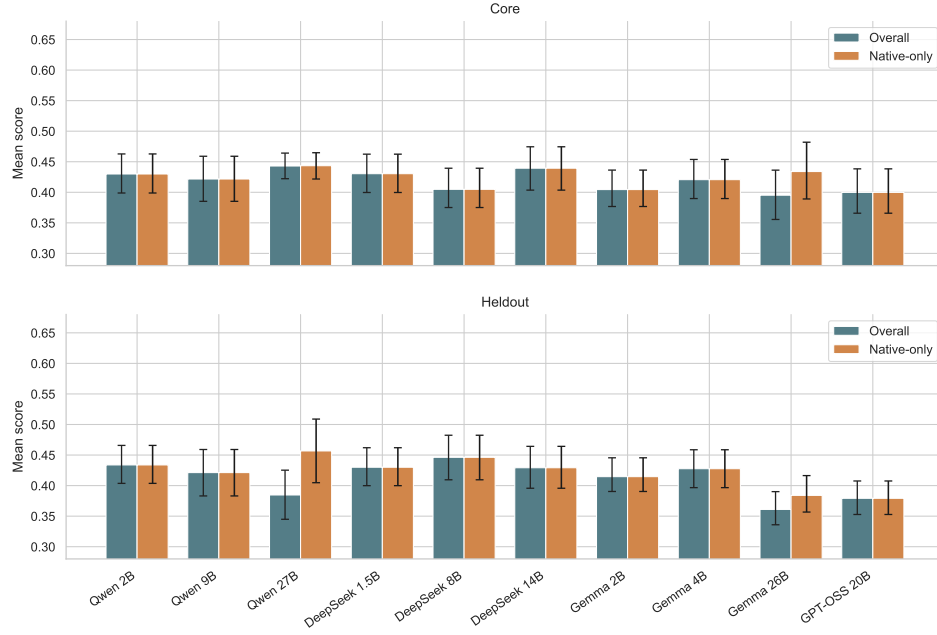


Figure 2: Observed score versus native-only score on the frozen release aggregate. Small-native-count estimates are diagnostic only, but the figure makes shell confounding visible instead of hiding it inside a single scalar mean.

Run / step	Public clue	Selected output fields	Strict / reward
DeepSeek 14B, P2 day 2	Day 2; 3 pending effects	initiative=ship_usage_pricing; message=retention_campaign; pricing=-0.1; support=incident_swarm	strict; -0.0302
DeepSeek 14B, P2 day 8	Day 8; 0 pending effects	initiative=refactor_incident_tooling; message=retention_campaign; pricing=0.0; support=incident_swarm	strict; 0.1427
Qwen 27B, P1 day 5	Day 5; 0 pending effects	{}	non-strict; -0.0216
Qwen 27B, P1 day 9	Day 9; 3 pending effects	initiative=launch_referral_loop; message=growth_push; pricing=0.0; support=balanced_triage	strict; -0.2013

Table 7: Exact selected fields copied from stored action payloads and episode logs in the frozen case studies. These are action outputs and scalar episode diagnostics, not chain-of-thought traces.

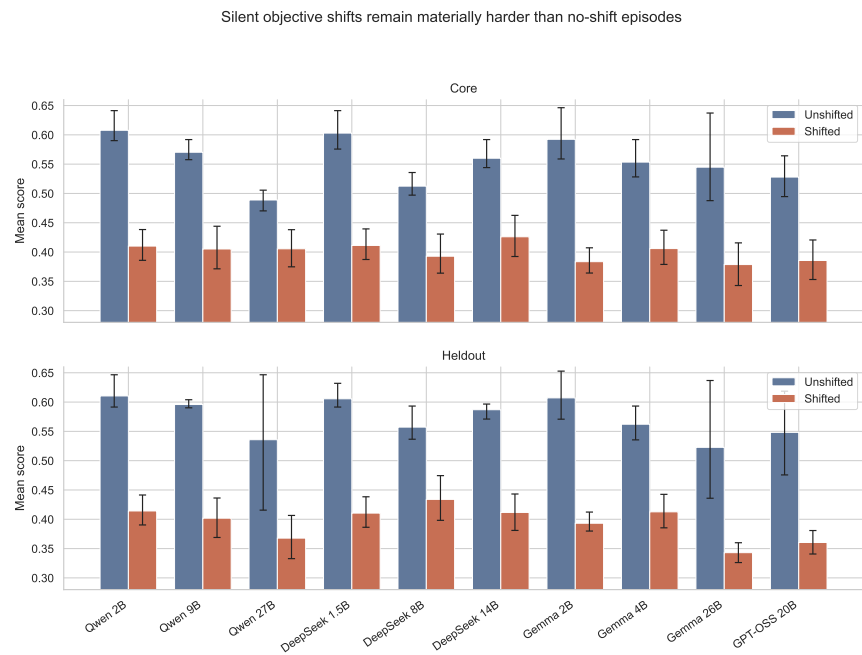


Figure 3: Shifted episodes remain materially harder than no-shift episodes on the frozen primary-task aggregate. Error bars denote 95% bootstrap confidence intervals for the score aggregates.

5 Discussion

LatentGoalOps clears the benchmark-science checks we care about most: baseline ordering, a persistent oracle gap, a consistent shift penalty, and no easy shortcut on the two anchor tasks. Because malformed actions change the trajectory itself, strict score, native-only score, and native episode rate need to be reported together.

6 Limitations and Intended Use

- The benchmark is synthetic and intended for comparative evaluation, not operational advice or direct real-world deployment claims.
- The heldout split preserves schema and grader semantics, and P3 is treated as shorter planning breadth rather than as a load-bearing anchor.
- GPT-OSS 120B is API-served, and this version does not include human baselines or a full human-audited grader validation study.
- The hosted-model probes are intentionally small because of API access, budget, and deadline constraints. They should be read as exploratory external probes, not exhaustive frontier-model coverage.
- The adaptation metric uses thresholded detection rules; we expose and audit those choices in the release protocol, but smoother alternatives remain possible.

7 Conclusion

LatentGoalOps isolates hidden-objective adaptation under silent non-stationarity. On the tighter P1/P2 anchor slice and the broader release aggregate, the benchmark shows no easy shortcut on the anchor tasks, a persistent shift penalty, and a clear gap between current local models and the myopic-oracle reference, while also revealing that some sharp heldout failures are interface-mediated. That combination argues for evaluating latent policy quality and action-shell reliability separately.

References

- AgentRewardBench Authors. Agentrewardbench: Evaluating automatic evaluators of web-agent trajectories. *arXiv preprint arXiv:2504.08942*, 2025. URL <https://arxiv.org/abs/2504.08942>.
- Alexandre Drouin, Maxime Gasse, Massimo Caccia, Issam H. Laradji, Manuel Del Verme, Tom Marty, Léo Boisvert, Megh Thakkar, Quentin Cappart, David Vazquez, Nicolas Chapados, and Alexandre Lacoste. Workarena: How capable are web agents at solving common knowledge work tasks? *arXiv preprint arXiv:2403.07718*, 2024. URL <https://arxiv.org/abs/2403.07718>.
- Saibo Geng, Hudson Cooper, Michał Moskal, Samuel Jenkins, Julian Berman, Nathan Ranchin, Robert West, Eric Horvitz, and Harsha Nori. Generating structured outputs from language models: Benchmark and studies. *arXiv preprint arXiv:2501.10868*, 2025. URL <https://arxiv.org/abs/2501.10868>.
- Thibault Le Sellier De Chezelles, Maxime Gasse, Alexandre Drouin, Massimo Caccia, Léo Boisvert, Megh Thakkar, Tom Marty, Rim Assouel, Sahar Omid Shayegan, Lawrence Keunho Jang, Xing Han Lü, Ori Yoran, Dehan Kong, Frank F. Xu, Siva Reddy, Quentin Cappart, Graham Neubig, Ruslan Salakhutdinov, Nicolas Chapados, and Alexandre Lacoste. Browsergym ecosystem for web agent research. *arXiv preprint arXiv:2412.05467*, 2024. URL <https://arxiv.org/abs/2412.05467>.
- Rachel Longjohn, Markelle Kelly, Sameer Singh, and Padhraic Smyth. Benchmark data repositories for better benchmarking. In *Advances in Neural Information Processing Systems 37: Datasets and Benchmarks Track*, 2024. URL https://papers.nips.cc/paper_files/paper/2024/hash/9d41fddf3169f4df3025fc60bfac94fa-Abstract-Datasets_and_Benchmarks_Track.html.

220 Mark Mazumder, Colby Banbury, Xiaozhe Yao, Bojan Karlas, William Gaviria Rojas, et al.
 221 Dataperf: Benchmarks for data-centric ai development. In *Advances in Neural Information*
 222 *Processing Systems 36: Datasets and Benchmarks Track*, 2023. URL [https://papers.nips.](https://papers.nips.cc/paper/2023/file/112db88215e25b3ae2750e9eefcded94-Paper-Datasets_and_Benchmarks.pdf)
 223 [cc/paper/2023/file/112db88215e25b3ae2750e9eefcded94-Paper-Datasets_and_](https://papers.nips.cc/paper/2023/file/112db88215e25b3ae2750e9eefcded94-Paper-Datasets_and_Benchmarks.pdf)
 224 [Benchmarks.pdf](https://papers.nips.cc/paper/2023/file/112db88215e25b3ae2750e9eefcded94-Paper-Datasets_and_Benchmarks.pdf).

225 Shishir G Patil, Huanzhi Mao, Fanjia Yan, Charlie Cheng-Jie Ji, Vishnu Suresh, Ion Stoica, and
 226 Joseph E. Gonzalez. The berkeley function calling leaderboard (bfcl): From tool use to agentic
 227 evaluation of large language models. In *Proceedings of the 42nd International Conference on*
 228 *Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 48371–
 229 48392, 2025. URL <https://proceedings.mlr.press/v267/patil25a.html>.

230 Anka Reuel, Alyssa Hardy, Jenna Wiens, Percy Liang, Matei Zaharia, Q. Vera Liao, Kacper Plawiak,
 231 Adrian Weller, et al. Betterbench: Assessing ai benchmarks, uncovering issues, and establishing
 232 best practices. *arXiv preprint arXiv:2411.12990*, 2024. URL [https://arxiv.org/abs/2411.](https://arxiv.org/abs/2411.12990)
 233 [12990](https://arxiv.org/abs/2411.12990).

234 Tianbao Xie, Danyang Zhang, Jixuan Chen, Xiaochuan Li, Siheng Zhao, Ruisheng Cao, Toh Jing
 235 Hua, Zhoujun Cheng, Dongchan Shin, Fangyu Lei, Yitao Liu, Yiheng Xu, Shuyan Zhou, Silvio
 236 Savarese, Caiming Xiong, Victor Zhong, and Tao Yu. Osworld: Benchmarking multimodal agents
 237 for open-ended tasks in real computer environments. *arXiv preprint arXiv:2404.07972*, 2024.
 238 URL <https://arxiv.org/abs/2404.07972>.

239 Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. τ -bench: A benchmark for
 240 tool-agent-user interaction in real-world domains. *arXiv preprint arXiv:2406.12045*, 2024. URL
 241 <https://arxiv.org/abs/2406.12045>.

242 Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng,
 243 Tianyue Ou, Yonatan Bisk, Daniel Fried, Uri Alon, and Graham Neubig. Webarena: A realistic
 244 web environment for building autonomous agents. *arXiv preprint arXiv:2307.13854*, 2023. URL
 245 <https://arxiv.org/abs/2307.13854>.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: The abstract and introduction state the benchmark contribution, the latent-adaptation scope, the heterogeneous scaling pattern, and the interface-reliability finding in a way that matches the reported results and stated caveats; see Abstract, Section 1, and Section 5.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [\[Yes\]](#)

Justification: The main paper contains a dedicated Limitations section covering provider heterogeneity, interface confounding, adaptation-metric sharpness, and synthetic-world scope; see Section 6.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [\[N/A\]](#)

Justification: The paper is an empirical benchmark paper and does not present formal theorems or proofs. The equations define benchmark mechanics and grading, not new theoretical results.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [\[Yes\]](#)

Justification: The manuscript and release bundle specify the frozen protocol, task scope, seed counts, split structure, model scope, regeneration commands, resolved manifest, and artifact layout; exploratory hosted API probes are labeled separately and their raw logs are included under `outputs/api-probe-2026-05-07`. See Sections 2–4 and the release files `paper/release/README.md` and `paper/release/REPRODUCTION.md`.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [\[Yes\]](#)

Justification: The submission now includes an anonymous reviewer archive that packages the runnable benchmark code, the frozen 2026-04-18 analysis root, the exact raw run directories referenced by `metadata/run_manifest.json`, the exploratory hosted API probe logs, and reproduction instructions; see Section 4 and the release files `paper/release/README.md`, `paper/release/REPRODUCTION.md`, and `paper/release/REVIEW_BUNDLE.md`.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [\[Yes\]](#)

Justification: The paper and release package specify the benchmark tasks, hidden-shift configuration, split structure, seed counts, scoring definitions, serving path, and model roles needed to interpret the evaluation; hosted API probes additionally disclose provider route, model identifiers, split coverage, and their exploratory status. See Sections 2–4 and the release protocol files.

298 **7. Experiment statistical significance**

299 Question: Does the paper report error bars suitably and correctly defined or other appropri-

300 ate information about the statistical significance of the experiments?

301 Answer: [Yes]

302 Justification: The main result and baseline tables report 95% bootstrap confidence inter-

303 vals on the headline aggregates, and the text explains the role of native-only estimates and

304 effective sample size; see Section 4 and the generated result tables.

305 **8. Experiments compute resources**

306 Question: For each experiment, does the paper provide sufficient information on the com-

307 puter resources (type of compute workers, memory, time of execution) needed to reproduce

308 the experiments?

309 Answer: [No]

310 Justification: The release package reports provider path, token counts, preserved wall-clock

311 timing, and a validation-host note in `paper/release/COMPUTE.md`, but the exact original

312 worker inventory for the full frozen runs was not preserved in the manifest. We therefore

313 keep the conservative No answer rather than over-claim hardware-normalized reproducibil-

314 ity.

315 **9. Code of ethics**

316 Question: Does the research conducted in the paper conform, in every respect, with the

317 NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

318 Answer: [Yes]

319 Justification: To the best of our knowledge, yes. The benchmark is synthetic, does

320 not involve human subjects, and the paper and datasheet explicitly restrict inappropriate

321 deployment-style uses; see Section 6 and `paper/release/DATASHEET.md`.

322 **10. Broader impacts**

323 Question: Does the paper discuss both potential positive societal impacts and negative

324 societal impacts of the work performed?

325 Answer: [Yes]

326 Justification: The release datasheet and the paper’s limitations discussion address appropri-

327 ate uses, inappropriate uses, and the risk of over-interpreting a synthetic benchmark as a

328 real deployment simulator; see Section 6 and `paper/release/DATASHEET.md`.

329 **11. Safeguards**

330 Question: Does the paper describe safeguards that have been put in place for responsible

331 release of data or models that have a high risk for misuse (e.g., pre-trained language models,

332 image generators, or scraped datasets)?

333 Answer: [N/A]

334 Justification: The paper releases a synthetic benchmark and frozen evaluation artifacts, not

335 a high-risk generative model or scraped dataset. The benchmark also includes intended-use

336 and limitation guidance in the release datasheet.

337 **12. Licenses for existing assets**

338 Question: Are the creators or original owners of assets (e.g., code, data, models), used in

339 the paper, properly credited and are the license and terms of use explicitly mentioned and

340 properly respected?

341 Answer: [Yes]

342 Justification: The release package now includes a consolidated asset-and-terms inventory

343 for the evaluated external model families and serving dependencies, with official source

344 links and notes on local alias handling; see `paper/release/THIRD_PARTY_ASSETS.md`.

345 **13. New assets**

346 Question: Are new assets introduced in the paper well documented and is the documenta-

347 tion provided alongside the assets?

348 Answer: [\[Yes\]](#)

349 Justification: The benchmark release includes documented code, a frozen artifact bundle,
 350 reproduction instructions, a datasheet, and Croissant metadata; see Section 4 and the files
 351 under `paper/release/`.

352 **14. Crowdsourcing and research with human subjects**

353 Question: For crowdsourcing experiments and research with human subjects, does the pa-
 354 per include the full text of instructions given to participants and screenshots, if applicable,
 355 as well as details about compensation (if any)?

356 Answer: [\[N/A\]](#)

357 Justification: The paper does not involve crowdsourcing or human-subject experiments.

358 **15. Institutional review board (IRB) approvals or equivalent for research with human**
 359 **subjects**

360 Question: Does the paper describe potential risks incurred by study participants, whether
 361 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)
 362 approvals (or an equivalent approval/review based on the requirements of your country or
 363 institution) were obtained?

364 Answer: [\[N/A\]](#)

365 Justification: The paper does not involve crowdsourcing or human-subject experiments, so
 366 IRB approval is not applicable.

367 **16. Declaration of LLM usage**

368 Question: Does the paper describe the usage of LLMs if it is an important, original, or
 369 non-standard component of the core methods in this research? Note that if the LLM is used
 370 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
 371 scientific rigor, or originality of the research, declaration is not required.

372 Answer: [\[Yes\]](#)

373 Justification: LLMs are the benchmarked systems in the experimental evaluation, including
 374 locally served models and hosted OpenRouter probes. The benchmark construction, deter-
 375 ministic scoring, and analysis pipeline do not rely on an auxiliary LLM-based labeling or
 376 generation procedure.